ACME: Agentic Contract-Based Multilevel Engineering for High-Assurance Complex Systems

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Abstract - Large language model (LLM) agents are increasingly employed to design complex systems, such as register-transfer level (RTL) hardware, controllers for teams of robots, and workflows of cooperating software agents. However, AI-generated designs are error-prone: agents decompose problems incorrectly, drift from their instructions over long tasks, and leave no record that allows an error to be traced back to its cause. Decomposing system-level requirements into component-level specifications is also hard in itself: distributed reactive synthesis is undecidable in general, and finding a decomposition whose components are realizable and whose composition is correct is still largely an art. In this project, we propose a framework, termed Agentic Contract-Based Multilevel Engineering (ACME), for sound and traceable decomposition of hardware, robotic, and agentic software systems based on assume-guarantee (A/G) contracts. Agents propose contracts at every level of a design hierarchy, and a contract engine with model checking and synthesis back ends decides whether each step is sound. ACME follows a three-pronged approach: an extension of contract theory to hardware, robotic, and agentic layers, including decidable classes of decomposition, sound approximations of the contract quotient, and circular reasoning; an iterative decomposition method that applies the same procedure to all three layers, optimized jointly with the cost of executing agent roles; and formal feedback that localizes failures, repairs decompositions, and grows a library of certified components reused across designs. ACME will be built on CHASE, our open-source contract-based design framework, and evaluated on hardware, multi-robot, and multi-agent software benchmarks.

Keywords – assume-guarantee contracts, compositional verification, reactive synthesis, GR(1), neurosymbolic reasoning, LLM agents, hardware formal verification, autoformalization.

Introduction

LLM agents are increasingly employed to design complex systems, from custom silicon to controllers for fleets of autonomous robots and workflows of cooperating software agents. Agents generate hardware assertions [1, 2], program invariants [3], and code at a scale no design team can review. However, AI-generated designs are error-prone, and the errors concentrate in decomposition, which every complex design requires. In our experience, agents that decompose end-to-end hardware properties may report timed-out runs as proofs or rely on circular assumptions across lemmas. Agents that delegate subtasks or compact their context restate the rules they must follow in natural language and drift away from them as the task proceeds [4]. Moreover, the reasoning behind a design is buried in natural-language transcripts, so an error found late cannot be traced back to the decision that caused it. As a result, AI-driven design of complex systems relies today on trial and error and on testing the final artifact. A new generation of design methods would be essential to make every step of AI-driven design traceable and checkable, and to keep the overall design correct as it is decomposed across components, agents, and abstraction levels.

Contract-based design offers a principled foundation for this goal. An A/G contract specifies what a component guarantees under assumptions on its environment [5]. A decomposition is correct when the composition of the component contracts refines the system contract, and the algebra of contracts provides operators, such as the quotient, that compute the weakest specification a missing component must satisfy [6, 7]. Contracts also give each step of a design a formal record that can be checked, stored, and re-

visited. Existing contract-based tools, including CHASE [8], Pacti [9], and OCRA [10], assume that contracts are written by human designers, and existing decomposition methods for synthesis rely on a fixed library of components [11]. Letting agents write the contracts raises several scientific questions. The existence of a decomposition into realizable components is undecidable for general architectures [12], so an agent may search indefinitely unless each failure is explained in a form that excludes whole families of candidates. The contract operators must also remain computable within the fragments that synthesis tools accept; for instance, the guarantee of the quotient of two GR(1) contracts contains the negation of assumptions and in general falls outside the GR(1) fragment.

In this project, we propose a framework, termed ACME, aiming to address these challenges by following a three-pronged approach that builds on our prior work. CHASE [8] provides a contract algebra with interfaces to verification and synthesis tools and to LLM agents. Our work on contract-based architecture exploration [13, 14] has shown that infeasibility certificates can be generalized via subgraph isomorphism to prune large design spaces, and received the Best Paper Award at DATE 2024. ACIES [15] uses LLM agents to decompose multi-robot requirements into GR(1) contracts that are certified and synthesized into controllers, and our ongoing work applies the same principle to RTL verification. If successful, this project will lay the foundations for AI-driven design of complex systems in which every decomposition proposed by an agent, for a hardware property, a robot mission, or a software task, is recorded as a set of contracts with a certificate that a solver can check, errors are traced to the step that caused them, and certified components are reused across designs.

Methods

ACME pursues three research thrusts. The first extends contract theory to the layers on which agents now operate; the second develops a single iterative decomposition method that applies to all of them; the third turns the outcome of each decomposition step into feedback, repairs, and a library of certified components that grows from one design to the next.

Thrust 1: Contract Theory for Hardware, Robotic, and Agentic Layers. Contract-based design has been developed mostly for cyber-physical components with fixed interfaces [5]. We will extend the modeling to the three layers targeted by this project. For hardware, a contract relates the ports of a register-transfer-level module over clock cycles and may refer to ghost state, such as the content of a buffer, which must itself be verified on the design. For robots, a contract couples a discrete mission layer, expressed in GR(1) [16], with continuous-time controllers whose guarantees come from control barrier functions [17] or Hamilton-Jacobi reachability. Learned policies, such as vision-language-action models [18], cannot be verified offline; their contracts are enforced at run time by safety filters and monitors, which report violated assumptions back to the decomposition. For agents, a contract in linear temporal logic on finite traces (LTL_f) [19] constrains the sequence of tool calls of a role, and its assumptions describe what the role may expect from the context it receives and from its sibling roles. On top of these models, we will study the algebraic questions that decomposition raises. The existence of a decomposition into realizable components is undecidable for general architectures [12]; we will identify classes of architectures and contracts for which it is decidable, starting from pipelines. The quotient of two GR(1) contracts in general leaves the fragment; we will develop approximations within GR(1) that strengthen the guarantees of the exact quotient, so that any controller synthesized for them is correct for the composition. Finally, circular dependencies arise naturally in pipelines, handshake protocols, and agents that wait on each other; we will adopt a circular A/G rule restricted to safety guarantees with an inductive side condition [20, 21], and characterize the decompositions it admits.

Thrust 2: Iterative Contract-Based Decomposition. ACME represents a design as a directed acyclic graph of contracts, built recursively. At an internal node, an agent proposes one contract for each subcomponent.

The proposal is accepted only if the component contracts are consistent, bounded checks on the actual components find no counterexample, and their composition refines the parent contract. Each subcomponent is then discharged by a back end or decomposed in turn, and a node that cannot satisfy its contract returns a witness to its parent, which revises the decomposition while certified subtrees are kept. The same procedure applies to all three layers, and only the leaves change: a hardware module that a model checker proves directly, a robot controller certified by a synthesis or reachability tool or a learned policy guarded at run time, or a task that one LLM agent completes within its context or a piece of deterministic code. In agentic workflows, a decomposition step corresponds to delegation: a parent agent passes to each role a task, a context, and a set of contracts, inherited or generated, and the quotient of the parent contract by the contracts of the sibling roles gives the weakest contract a role must satisfy. A compacted context is accepted only if its contracts refine those of the full context, and the choice between one agent and several [22] is cast as contract-based architecture exploration [13, 14]. Because the contracts of a role determine which tools it may call next, the decomposition can be optimized jointly with the cost of executing the roles, by partitioning the key-value cache of the model into modules aligned with contract states [23, 24] and prefetching modules for reachable states.

Thrust 3: Formal Feedback, Repair, and Reuse. The value of the decomposition engine for an agent lies in what it returns when a step fails. The engine separates a formalism-independent contract algebra, checks reduced to validity or satisfiability, and formalism-specific analyses such as unrealizable cores, vacuity [25], and well-separation [26]; every failed check returns a counterexample or a minimal core that points to the responsible terms. A Model Context Protocol (MCP) server and an agent skill expose these operations to agents, while no LLM is part of the engine. Refinement failures are first repaired with the quotient, before any agent call. As learned clauses prune the search of a SAT solver, the core of a failed decomposition is generalized via subgraph isomorphism [13] into a constraint that excludes every candidate assigning an isomorphic component the same terms. Certified contracts are stored together with their components, so that the contract library grows with each design and a component can be reused elsewhere after its assumptions are rechecked; the coverage of the library is measured by injecting faults into the components.

Evaluation and preliminary results. In hardware, component contracts are SystemVerilog assertions (SVAs) with ghost-state models, verified with Cadence Jasper and the open-source SymbiYosys. Our prototype certified an end-to-end in-order delivery property of a HierSVA asynchronous link with 25 Jasper runs and about 16,000 LLM input tokens, which flat Jasper did not decide in the same time; the decomposition found at FIFO depth 8 certified depths 16 to 512 with no further LLM calls, and an agent without the protocol used 24 times as many tokens without completing the proof. In multi-robot control, ACIES [15] decomposes mission requirements into GR(1) contracts, synthesizes controllers, and executes them under control barrier functions with runtime monitors; in a three-level decomposition with 42 nodes, the contract library closed 26 of the 39 leaves. For agentic software, we will add delegation, compaction, and cache-aware tool exposure one at a time to a single-agent executor on long-horizon tasks from AppWorld [27], against summarization-based compaction.

Expected results

Months	Research result	Deliverable
1–3	Contract models for the three layers; quotient approximations within GR(1) (Thrust 1)	Technical report; CHASE release with MCP server and agent skill
4–6	Circular A/G rule with a soundness proof (Thrust 1); layered feedback and learned constraints (Thrust 3)	Open-source hardware flow with SymbiYosys on AWS; RTL benchmark with faulty variants

Months	Research result	Deliverable
7–9	Iterative decomposition across layers; contract-based delegation (Thrust 2)	Hardware and multi-robot evaluation; conference paper
10–12	Joint optimization with inference cost (Thrust 2); certified library and reuse (Thrust 3)	Agent evaluation; tool paper on CHASE; talk to Amazon teams

Beyond the first year, we expect these results to extend to contracts with probabilistic guarantees and to design flows in which the three domains share certified components. All code, benchmarks, and run traces will be released under open-source licenses.

Funds needed

Cash funding ($80,000–$100,000). Partial support of one graduate student researcher working on the theory, the framework, and the benchmarks (stipend and fees, [amount]), and travel to present the results at conferences ([amount]). The upper end of the range would add partial support of a second student working on Thrust 2. No indirect expenses are included.

AWS Promotional Credits ($25,000–$40,000). Amazon Bedrock for agent inference, since our hardware and robotic pipelines already invoke Claude models through Bedrock; Amazon EC2 compute-optimized instances for parallel runs of nuXmv, slugs, and SymbiYosys; and Amazon S3 for benchmarks and run traces. Jasper runs use campus licenses.

Additional information

1. Application domain and analysis techniques. Compositional design of hardware, multi-robot systems, and LLM agents, using A/G contracts, linear temporal logic (LTL) and LTL_f model checking (nuXmv), satisfiability modulo theories (SMT) reasoning, GR(1) synthesis (slugs), and RTL property verification (Jasper, SymbiYosys).

2. Current applications. CHASE has been used for requirement formalization and validation of cyber-physical systems [8], contract-based architecture exploration [13, 14], and consistency and availability analysis of distributed systems in collaboration with Lockheed Martin [28]. Our RTL flow runs on OpenTitan and HierSVA designs.

3. Potential applications to Amazon. Verification of AWS silicon; controllers for Amazon Robotics fleets; multi-agent workflows on Amazon Bedrock whose obligations are preserved across delegation, at lower serving cost; and consistency and vacuity analysis of formalized policies, complementary to Automated Reasoning checks in Bedrock Guardrails.

4–5. Prior Amazon Research Awards. No. The project team has not received Amazon cash funding or AWS Promotional Credits in the past three years.

6. Assumptions and soundness. A certified result is sound when (i) the back ends return correct verdicts; (ii) the translation into each back end preserves the semantics of the contracts; (iii) every composition step is acyclic or satisfies the side condition of the circular rule in Thrust 1; and (iv) the top-level specification is correct, which is why agent-written contracts are checked for consistency and vacuity. Bounded results are never counted as proofs, LLM outputs are never trusted, and quotient approximations are used only in the direction that preserves refinement.

7. Tools. *License:* BSD 3-Clause for CHASE and its MCP server, Apache 2.0 for the RTL flow, and MIT for ACIES. *On-boarding:* a website with a usage guide and API reference, a Docker image, and the agent skill; a tutorial will be added. *Maintenance:* over 90 commits in the last three months; one active contributor.

Appendix A – Reference

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